Block 4 – Session 2

Cloud Computing & Cloud Networking

Arquitectura de Xarxes

Universitat Pompeu Fabra

1. From Virtualization to Cloud
2. Fundamentals of Cloud Computing
3. From VLANs to Overlay Networks
4. Cloud Network Architecture
5. Cloud Governance, Sovereignty & Compliance
6. Session Summary

Outline

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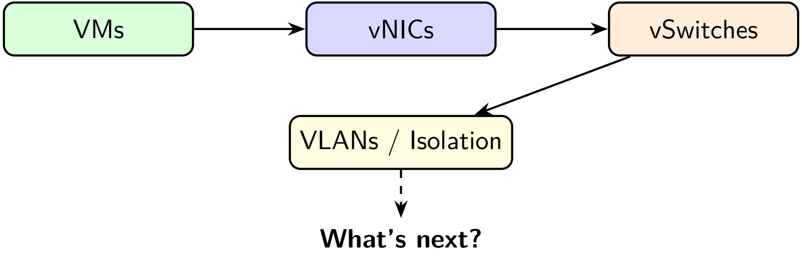
If We Can Virtualize Everything... Why Not Sell It?

Virtualization is the technology.
Cloud computing is the business model.

Learning objectives:

1. Understand cloud computing's definition and essential characteristics
2. Distinguish IaaS, PaaS, and SaaS service models
3. Design a basic cloud network with VPCs, subnets, and security controls

Recap – What We Virtualized (Session 1)



- Compute virtualization: VMs and containers
- Network virtualization: vNICs, vSwitches, vRouters, VXLAN overlays
- Together: the **building blocks** of the cloud

The Leap to Cloud

- Virtualization = technology (**how** you run multiple workloads)
- Cloud = business model (**what** you sell to customers)

Virtualization	**Cloud Computing**
Run multiple VMs on one host	Offer compute as a **service**
Internal IT optimization	External/internal consumption
Manual provisioning possible	**Automated, self-service**
You own the hardware	Provider owns the hardware

Key difference: **automation and self-service** (no phone calls, no tickets, no waiting).

Discussion: From Virtualization to Cloud

Your company already runs VMs on its own servers. Why would you move to the cloud instead of keeping everything in-house?

Hints: think about CapEx vs OpEx, scaling speed, and who manages what.

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Cloud Computing – Definition (NIST)

A model for enabling ubiquitous, convenient, ***on-demand network access*** *to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort.*

Source: NIST SP 800-145 (2011). Still the standard definition used industry-wide.

Five Essential Characteristics

Characteristic	What it means
On-demand self-service	Provision resources without human interaction
Broad network access	Available over the network from any device
Resource pooling	Shared infrastructure, multi-tenant
Rapid elasticity	Scale up/down automatically
Measured service	Pay only for what you use (metered)

- **All five** must be present to qualify as "cloud computing"
- If you need to call someone to get a VM $\rightarrow$ it is not cloud

Source: NIST SP 800-145, Mell & Grance, 2011.

On-Demand Self-Service

- Users provision resources through a **portal or API**
- No phone calls, no tickets, no waiting for approval
- Example: launch a VM in **30 seconds** via a web console

Analogy: vending machine (cloud) vs restaurant with a waiter (traditional IT).

Broad Network Access + Resource Pooling

Broad network access:

- Resources accessible from **any device** with network connectivity
- Standard protocols: HTTPS, SSH, REST APIs
- No proprietary client software needed

Resource pooling:

- Provider's resources serve **multiple tenants** simultaneously
- Tenants do not know (or care) where their resources are physically located
- Dynamically assigned based on demand
- Builds on the **multi-tenancy** concepts from Session 1

Rapid Elasticity + Measured Service

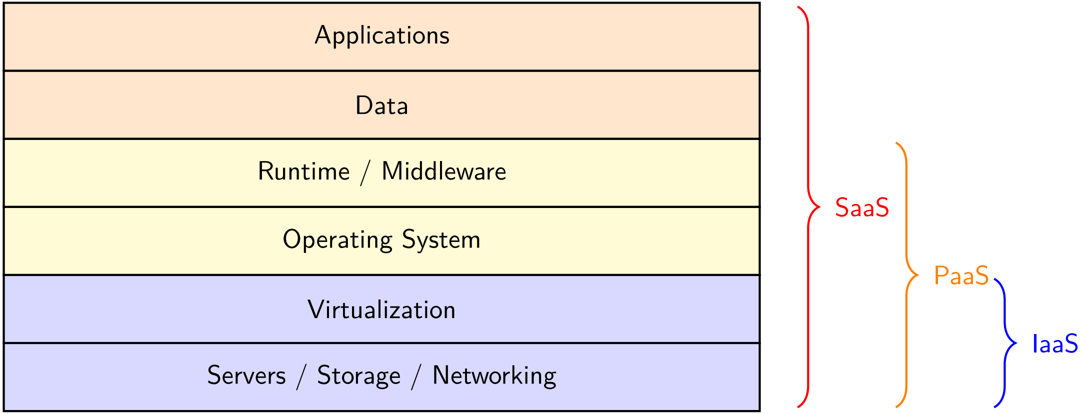
Rapid elasticity:

- Resources appear **unlimited** to the consumer
- Scale out (add) and scale in (remove) automatically
- Example: web store on Black Friday → 10× servers, then back to normal

Measured service:

- Usage is **metered** (CPU-hours, GB stored, GB transferred)
- **Pay-per-use** model → OpEx instead of CapEx
- Transparent monitoring and billing reports

Cloud Service Models – The Stack



IaaS – Infrastructure as a Service

- Provider gives you: **VMs, storage, networks**
- You manage: OS, middleware, applications, data
- **Maximum control, maximum responsibility**

Examples:

- **AWS EC2**: virtual machines
- **Azure Virtual Machines**
- **Google Compute Engine**

Use case: "I want a Linux server in Frankfurt, 4 CPUs, 16 GB RAM, right now."

Source: AWS EC2 Documentation, docs.aws.amazon.com.

PaaS – Platform as a Service

- Provider gives you: **runtime environment + tools**
- You manage: only your **application code and data**
- No need to manage OS, patches, or scaling infrastructure

Examples:

- **Google App Engine**
- **AWS Elastic Beanstalk**
- **Heroku**

Use case: "I wrote a Python web app. Just run it, scale it, I do not care about servers."

SaaS – Software as a Service

- Provider gives you: a **complete application**
- You manage: your **data and user configuration**
- No installation, no maintenance, no updates to worry about

Examples:

- **Google Workspace** (Gmail, Docs, Drive)
- **Microsoft 365** (Word, Excel, Teams)
- **Salesforce**, **Slack**, **Zoom**

Use case: "I need email for 500 employees. I do not want to run a mail server."

Service Models – Who Manages What?

Component	IaaS	PaaS	SaaS
Applications	You	You	Provider
Data	You	You	You
Runtime / Middleware	You	Provider	Provider
Operating System	You	Provider	Provider
Virtualization	Provider	Provider	Provider
Hardware	Provider	Provider	Provider

- IaaS $\rightarrow$ SaaS: **less control, less responsibility**
- Major providers (AWS, Azure, GCP) offer all three models

Discussion: Cloud Fundamentals

A university wants to offer a web-based tool for students to submit assignments. Should they choose IaaS, PaaS, or SaaS? Why?

Hints: consider the IT team size, maintenance burden, and how much control they need.

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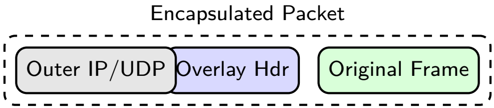
The VLAN Scalability Problem

- In Session 1 we used **VLANs** to isolate tenants on virtual networks
- VLANs use a **12-bit ID** → maximum **4,096** virtual networks
- Cloud providers host **millions** of tenants → VLANs are not enough
- Additional limitations:
  - VLANs are confined to a **single L2 domain**
  - Moving a VM to another host may require VLAN reconfiguration

We need a technology that scales beyond 4K networks and works across L3 boundaries.

Overlay Networks: Concept

- An **overlay network** is a virtual network built **on top of** the physical one
- Uses **encapsulation**: wrap virtual frames inside physical packets



- The physical network only sees the **outer headers**
- The virtual (tenant) traffic is hidden inside → **full isolation**

VXLAN: Overview

- **VXLAN** (Virtual eXtensible LAN): most widely used overlay protocol
- Encapsulates L2 frames in **UDP packets** over the physical network
- Uses a **24-bit VNI** (VXLAN Network Identifier)
  - 2^{24} = **16 million** virtual networks (vs 4,096 VLANs)
- Endpoints: **VTEPs** (VXLAN Tunnel Endpoints)
  - Encapsulate at source, decapsulate at destination

Source: IETF RFC 7348, "Virtual eXtensible Local Area Network (VXLAN)," 2014.

VXLAN: How It Works

- VM A sends a frame $\rightarrow$ VTEP 1 wraps it in UDP with a VNI
- Travels over the physical network as a regular UDP packet
- VTEP 2 strips outer headers, delivers original frame to VM B
- VMs on the **same VNI** form an isolated L2 domain

Overlay Benefits for Scalability

- **16 million virtual networks** (vs 4,096 VLANs)
- Physical network does not need to know about tenants
- VMs can **migrate** across hosts without reconfiguring the underlay
- Decouples **virtual topology** from physical topology

This is the foundation of cloud networking. Next: how providers build **VPCs** on top of overlays.

Source: IETF RFC 8926, "Geneve: Generic Network Virtualization Encapsulation," 2020.

Discussion: Overlay Networks

Why can the physical network remain simple
if we use overlay networks?
What are the trade-offs of encapsulation?

Hints: think about overhead (extra headers), MTU implications, and troubleshooting complexity.

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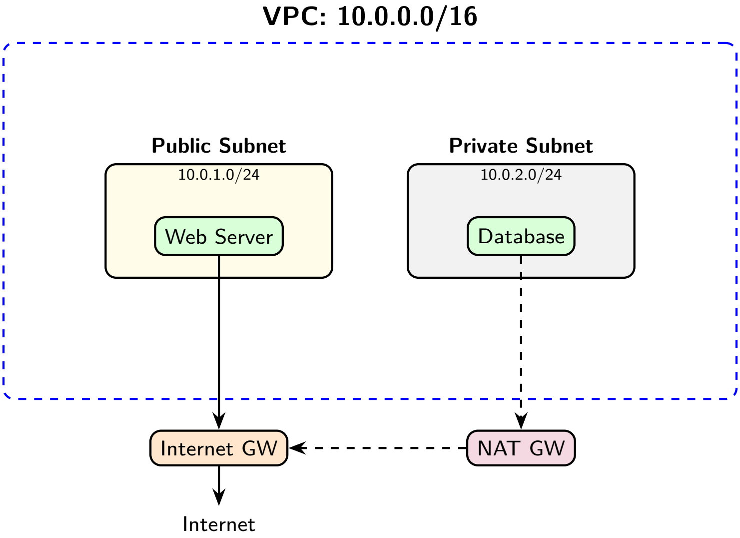
Virtual Private Cloud (VPC)

- A **VPC** is your own **isolated virtual network** inside a cloud provider
- Think of it as your private data center in the cloud
- You define:
  - **Address space** (e.g., `10.0.0.0/16`)
  - **Subnets** (subdivisions of the address space)
  - **Routing rules** and **security policies**

Key: the VPC is built on top of the **overlay network** concepts from Session 1.

Source: AWS, "Amazon VPC User Guide," docs.aws.amazon.com.

VPC Architecture



Subnets – Public vs Private

Public subnet:

- Has a route to an **Internet Gateway**
- Instances can have **public IP addresses**
- Accessible from the Internet (if security rules allow)
- Use for: web servers, load balancers, bastion hosts

Private subnet:

- **No direct route** to the Internet
- Instances only have **private IPs**
- Can access Internet outbound via **NAT Gateway**
- Use for: databases, internal services, application backends

Route Tables

- Each subnet is associated with a **route table**
- Route table = set of rules determining where traffic goes

Destination	**Target**	**Subnet type**
`10.0.0.0/16`	local (within VPC)	Both
`0.0.0.0/0`	Internet Gateway	Public
`0.0.0.0/0`	NAT Gateway	Private

- **Local route**: traffic within the VPC stays inside (always present)
- **Default route** (`0.0.0.0/0`): where to send everything else
- Public subnets → IGW; Private subnets → NAT GW

Internet Gateway vs NAT Gateway

	Internet Gateway	**NAT Gateway**
Direction	Bidirectional	Outbound only
Use case	Public-facing services	Private instances need updates
Assigned to	VPC (one per VPC)	Public subnet
Instance needs	Public IP	Only private IP

- Internet GW: allows the world to **reach** your public instances
- NAT GW: lets private instances **reach out** without being reachable
- Same **NAT concept from Block 3**, now as a managed cloud service

Security Groups – Instance-Level Firewall

- **Security group** = virtual firewall attached to an instance
- Rules specify allowed **inbound** and **outbound** traffic
- **Stateful**: allow inbound → response automatically allowed

Direction	Protocol	Port	Source/Dest
Inbound	TCP	80	`0.0.0.0/0`
Inbound	TCP	443	`0.0.0.0/0`
Inbound	TCP	22	`10.0.0.0/16`
Outbound	All	All	`0.0.0.0/0`

Source: AWS, "Security Groups for Your VPC," docs.aws.amazon.com.

Network ACLs – Subnet-Level Firewall

- **Network ACL** = firewall at the **subnet** level
- Rules evaluated in **order** (numbered, first match wins)
- **Stateless**: must explicitly allow both inbound AND outbound

Rule #	Direction	Protocol	Port	Action
100	Inbound	TCP	80	ALLOW
200	Inbound	TCP	443	ALLOW
*	Inbound	All	All	DENY

Source: AWS, "Network ACLs," docs.aws.amazon.com.

Security Groups vs Network ACLs

	Security Groups	**Network ACLs**
Scope	Instance level	Subnet level
State	Stateful	Stateless
Rules	Allow only	Allow + Deny
Evaluation	All rules evaluated	First match wins
Default	Deny all inbound	Allow all

Defense in depth: use both together.

- Security Groups → fine-grained per-instance control
- Network ACLs → broad subnet-level guardrails

Discussion: Cloud Networking

You deploy a web application in a VPC.
The web server needs to be public, but the database must not be reachable from the Internet. How?

Hints: think about public vs private subnets, security groups, and NAT gateways.

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Cloud Governance

- **Governance** = policies and controls for managing cloud resources
- Key areas:
  - **Resource management**: who can create/delete what, tagging
  - **Cost control**: budgets, alerts, spending limits
  - **Access control**: Identity and Access Management (IAM)
  - **Compliance**: ensuring regulatory requirements are met

Without governance: resource sprawl, security gaps, surprise bills.

Data Sovereignty and GDPR

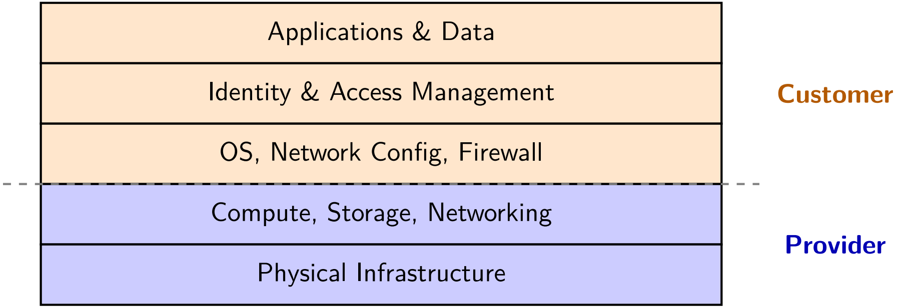
- **Data sovereignty**: data is subject to the laws where it is **stored**
- Cloud providers have data centers worldwide → your data location matters
- Risks: jurisdiction conflicts (EU data in US), government access (CLOUD Act)

GDPR (EU, 2018) – key principles:

- **Lawfulness** + **purpose limitation** + **data minimization**
- **Right to erasure**: users can request data deletion
- **Breach notification**: 72 hours to report
- Relevance: you must know **where** your data is and **who** can access it

Source: Regulation (EU) 2016/679, General Data Protection Regulation, 2018.

Shared Responsibility Model



- **Provider**: physical security, hardware, hypervisors, network fabric
- **Customer**: data, access policies, OS patches, application security
- The dividing line **shifts** depending on IaaS / PaaS / SaaS

Source: AWS, "Shared Responsibility Model," aws.amazon.com/compliance/shared-responsibility-model.

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Key Takeaways

1. Virtualization is the **technology**; cloud is the **business model**
2. Five NIST characteristics define true cloud computing
3. **IaaS** / **PaaS** / **SaaS** differ in who manages what
4. **VXLAN** overlays scale isolation to 16M networks (vs 4K VLANs)
5. A **VPC** is your isolated cloud network, built on top of overlays
6. **Security groups** (instance) + **ACLs** (subnet) = defense in depth
7. **Data sovereignty** and GDPR affect where and how you store data

Discussion

A Spanish startup stores user data in an AWS region in Ireland. A US law enforcement agency requests access. Who is responsible? What does GDPR say?

Think about: shared responsibility, data location, and which laws apply.

References

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